

LIGHTS, CAMERA, PROFIT

Using Data Analytics to Build an
Oscar-Winning Movie Studio

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BUSINESS SCENARIO

The Challenge

We were briefed by two stakeholders of a new movie studio:

#1



The Investor

- Maximize financial returns
- Budget of approximately \$1 million
- What are the odds of creating a profitable movie?

#2



The Producer

- Wants to win Academy Awards
- Wants to know which directors and genres have the highest Oscar potential
- Needs guidance on how much should be invested in an Oscar-calibre film

Our goal

Use historical movie data to identify patterns that increase the likelihood of financial success and Oscar recognition.

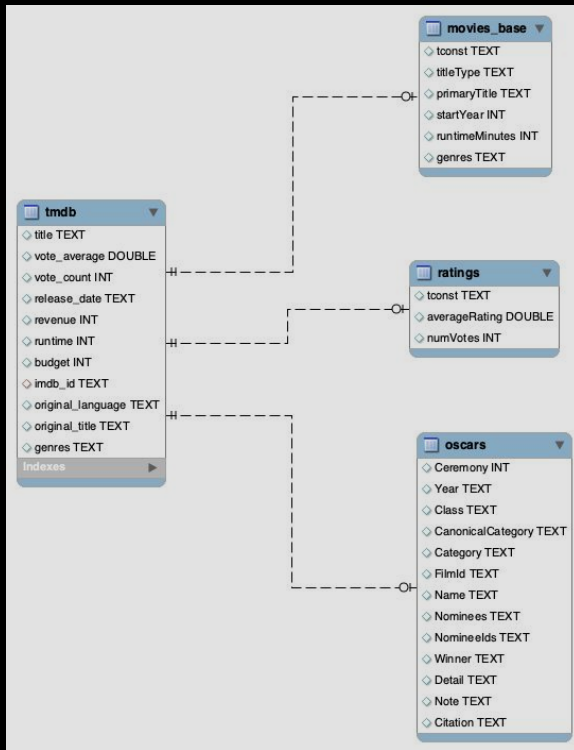
Project Overview

Data sources

- TMDb Dataset: Movie budgets, Revenue, Genres
- Movie base Dataset: Title, Year, Runtime, Genres
- IMDb Ratings Dataset: Audience ratings
- Oscars Dataset: Nominees, Winners, Award categories

Time period

- Movies released between 2000 and 2025



METHODOLOGY

01

- Cleaned and imported TMDb, IMDb and Oscars datasets into MySQL

02

- Calculated profitability metrics: Profit and Return on Investment

03

- Aggregated and compared Budgets, Revenue, Ratings, Genres, Directors

04

- Created visualisations to display the results

```
CREATE VIEW directing_nominees AS
SELECT
Year,
Film,
Nominees,
Winner
FROM oscars
WHERE Category = 'DIRECTING'
AND Year BETWEEN 2000 AND 2026;
```

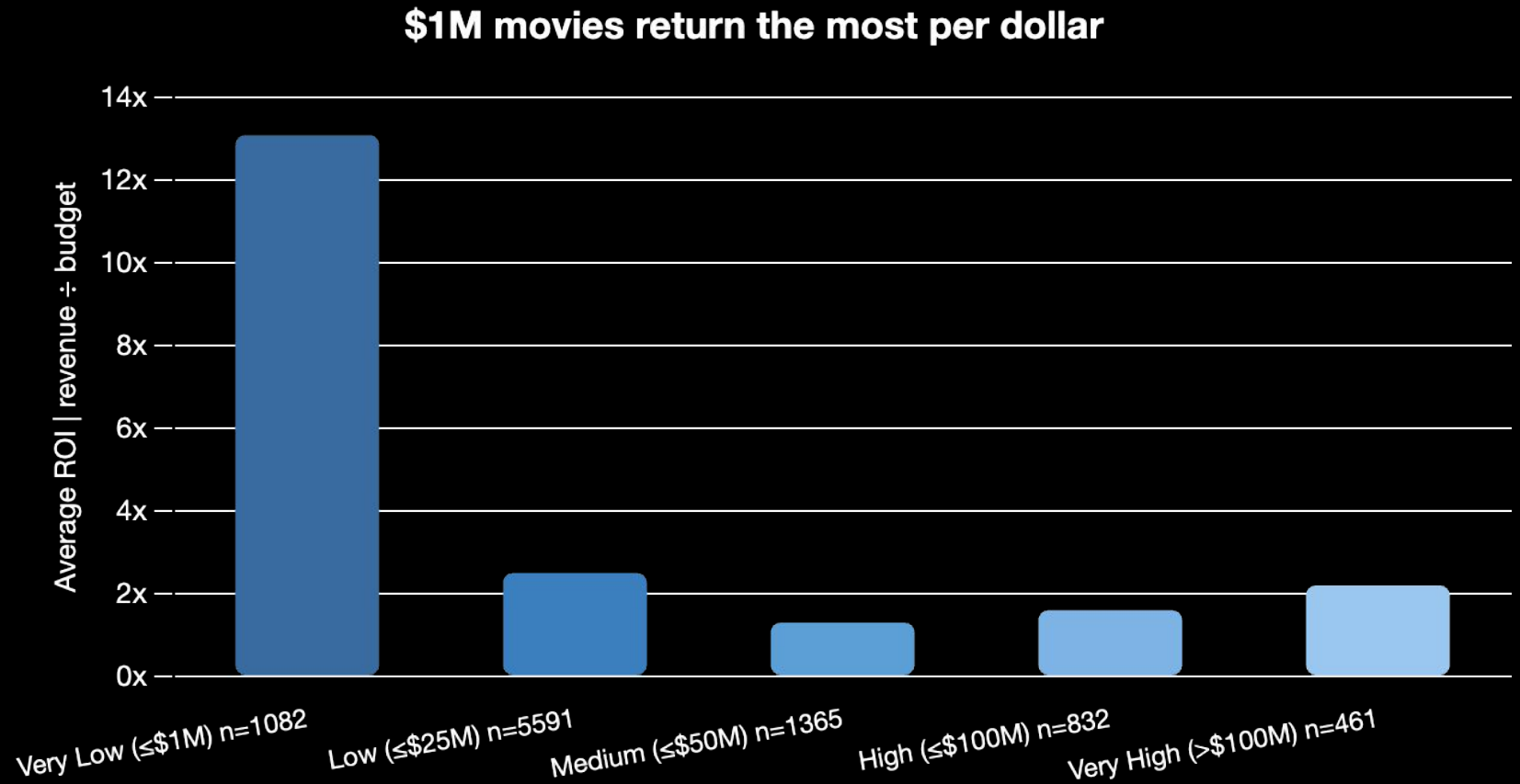
```
-- Create a view by joining the nominees to the movie financials
CREATE VIEW directing_financials AS
SELECT
directing_nominees.Nominees,
directing_nominees.Film,
directing_nominees.Winner,
tmdb.budget,
tmdb.revenue,
tmdb.release_date,
((tmdb.revenue - tmdb.budget) / tmdb.budget) * 100 AS roi
FROM directing_nominees
JOIN tmdb
ON directing_nominees.Film = tmdb.title
WHERE tmdb.budget > 10000
AND YEAR(tmdb.release_date) BETWEEN 2000 AND 2026
AND tmdb.revenue > 0;
```



BUSINESS QUESTION: BUDGET VS ROI



Can a \$1 million movie be financially successful?

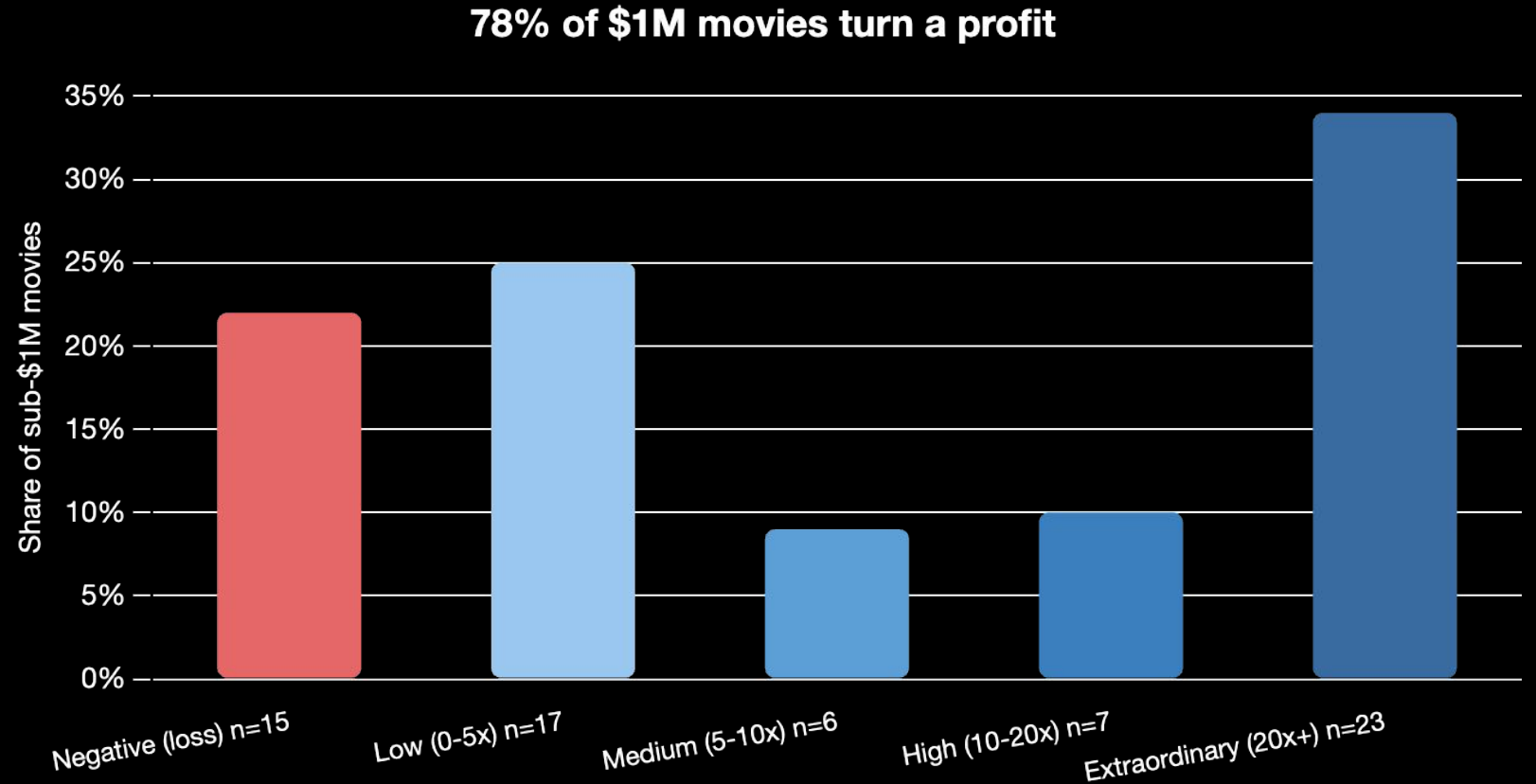




BUSINESS QUESTION: RISK VS REWARD



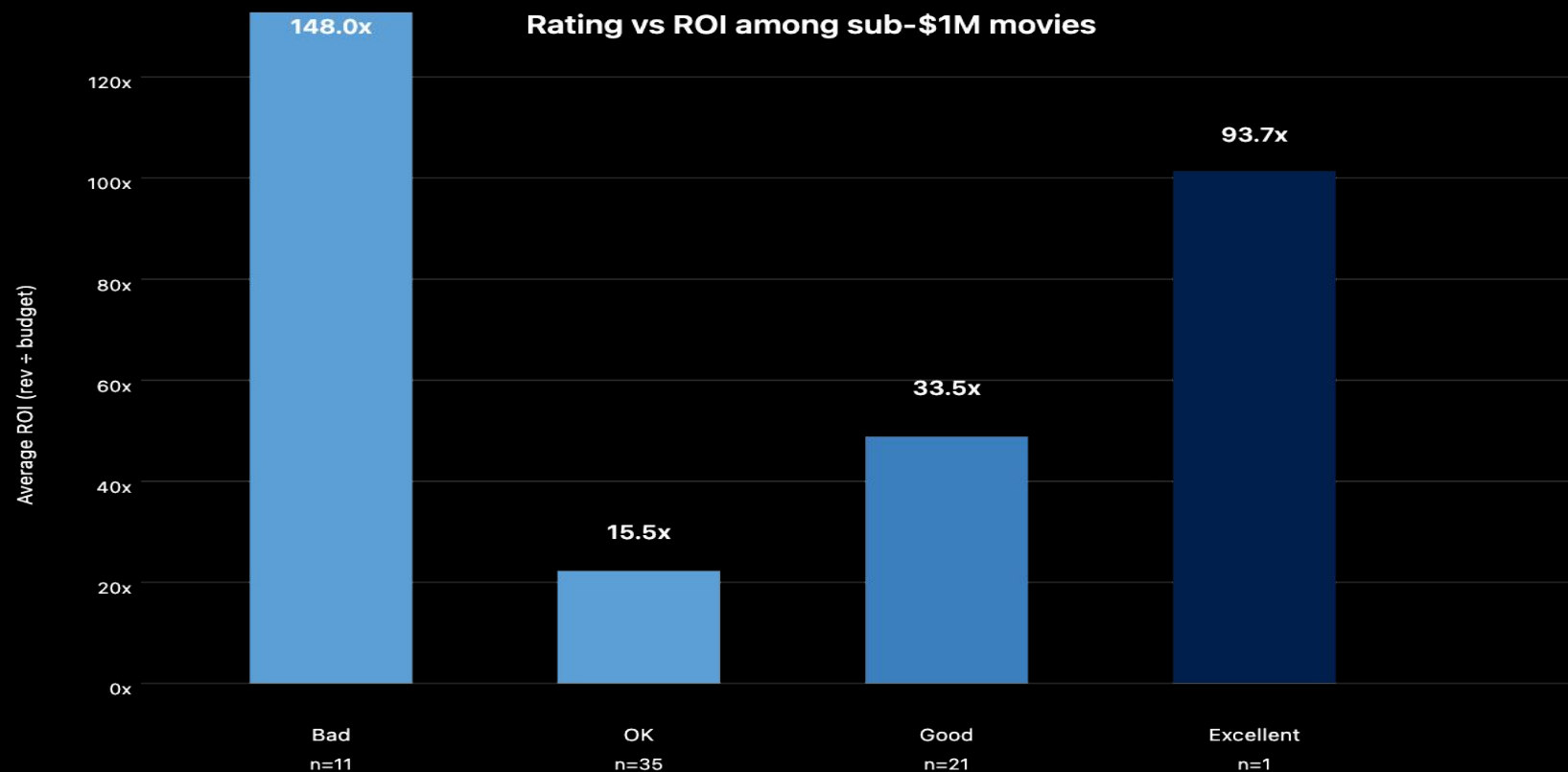
What are the chances of a very low budget movie earning a high ROI?





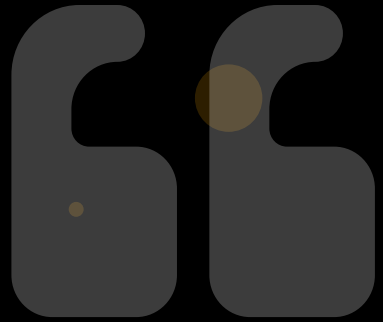
BUSINESS QUESTION: RATINGS VS ROI

Do higher-rated movies earn more money?

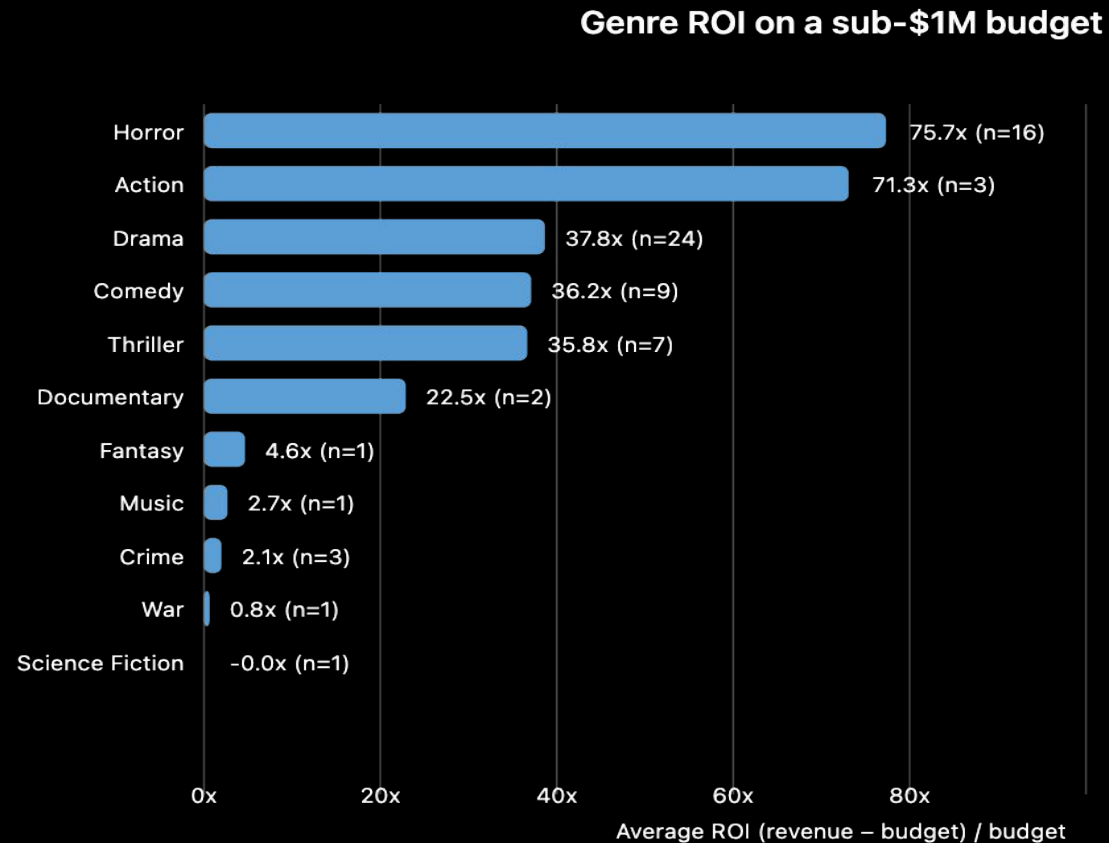




BUSINESS QUESTION: GENRE VS ROI

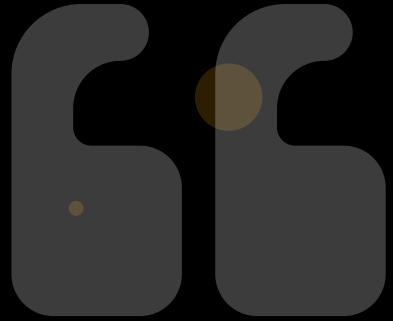


Which genre
earns the
highest ROI?

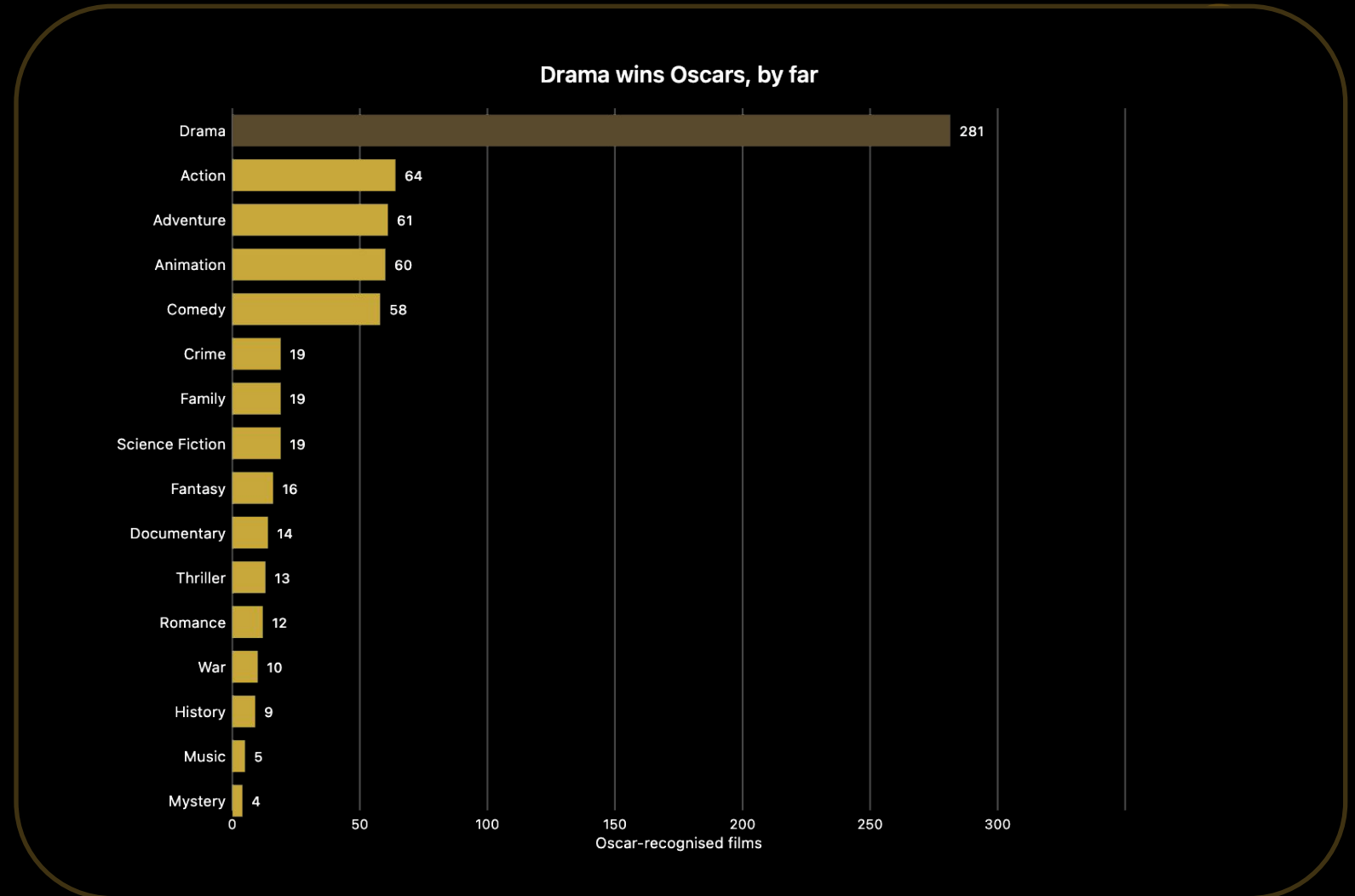




BUSINESS QUESTION: OSCARS VS. GENRE



Which genres
win the most
Oscars?



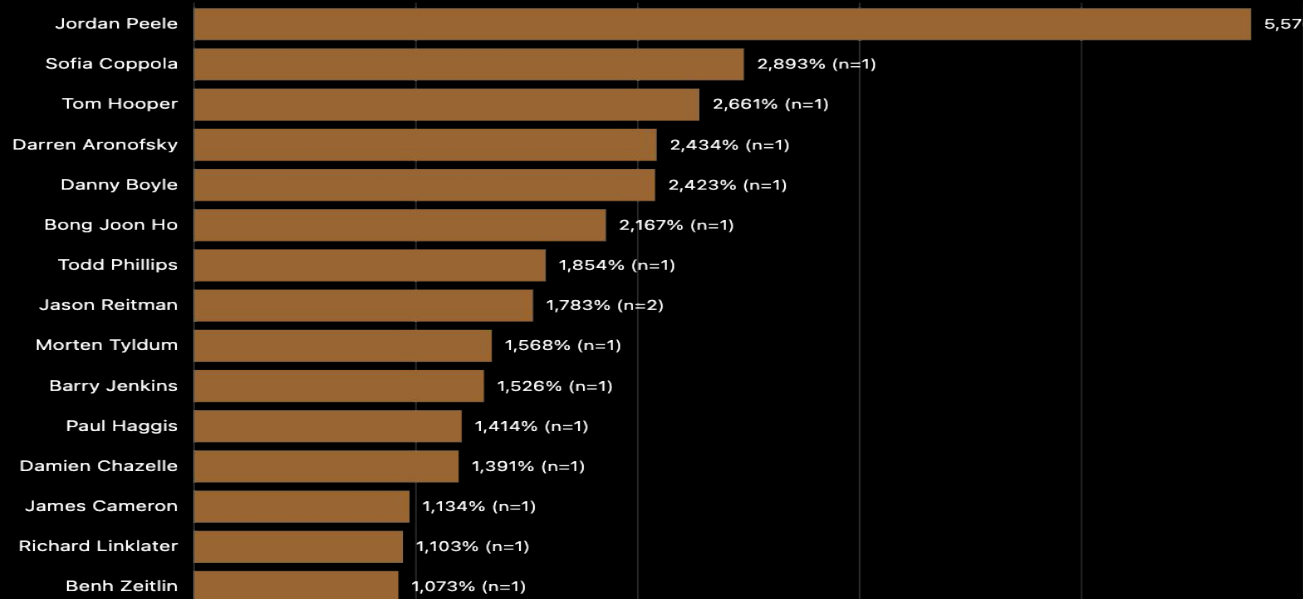


BUSINESS QUESTION: DIRECTORS VS ROI

Which nominated directors have the highest ROI?

99

Top 15 nominated directors by average ROI



Top 10 directors with more than 1 Oscar nomination or win

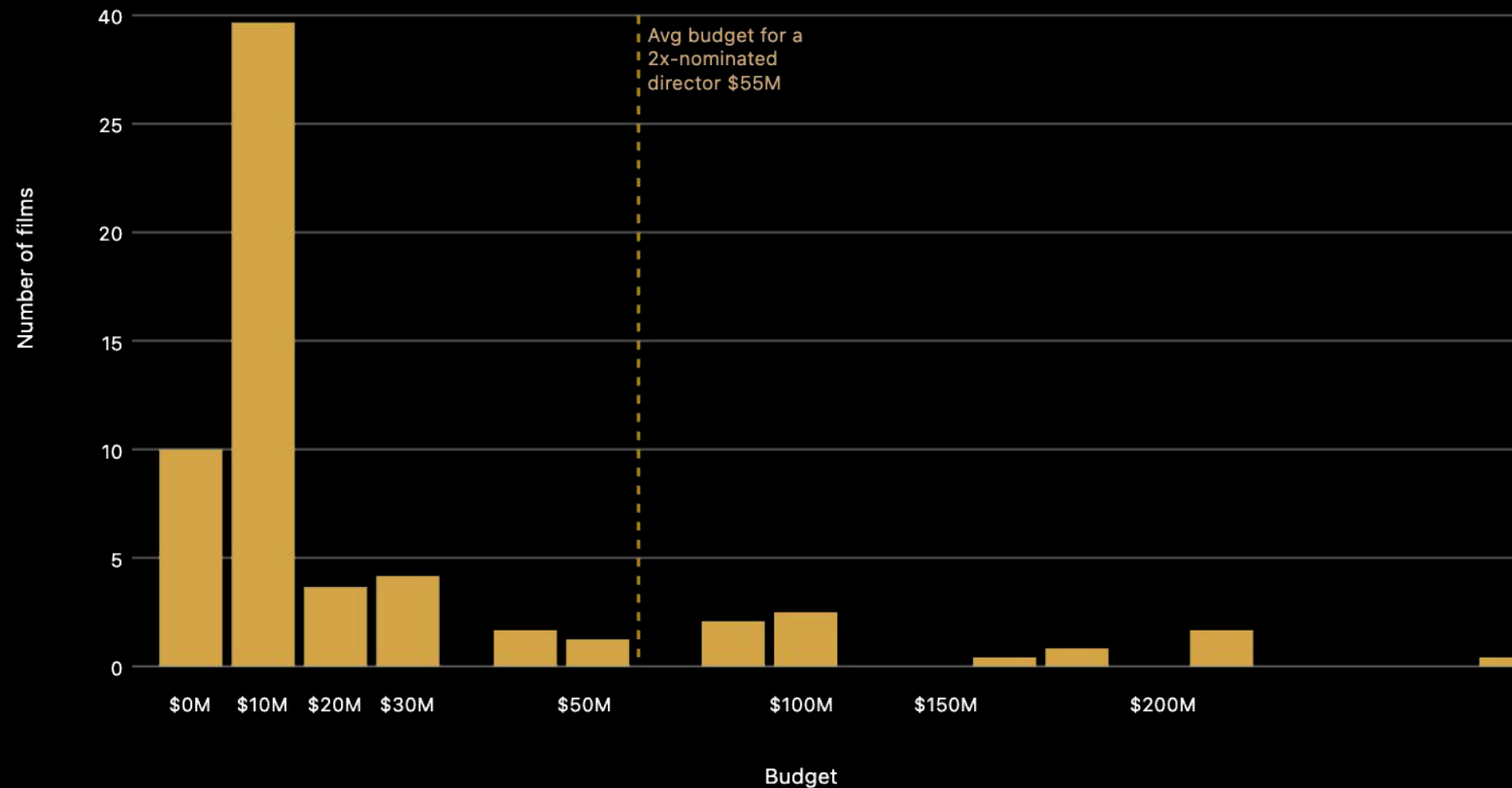
Nominees	movies	avg_revenue	avg_roi	avg_budget
Jason Reitman	2	199607710.00	1782.84	16250000.00
Peter Jackson	2	995128671.50	963.64	93500000.00
Ang Lee	3	333528687.33	911.76	50333333.33
Stephen Daldry	3	86593526.33	797.44	20666666.67
David O. Russell	3	193733756.33	609.39	28666666.67
Joel Coen	2	211952046.50	575.20	31500000.00
Ethan Coen	2	730350000.00	542.52	125000000.00
Christopher Nolan	2	730350000.00	542.52	125000000.00
Alexander Payne	3	104809453.33	500.43	16000000.00
Adam McKay	3	70280193.33	495.62	29373333.33
Clint Eastwood	3	147419631.33	470.43	24666666.67



BUSINESS QUESTION: DIRECTORS VS BUDGET

What average budget should we expect to invest in these filmmakers?

Oscar-nominated films typically cost tens of millions



Executive Recommendations



The Producer

#1

Prioritize drama-focused projects

#2

Work with proven Oscar-nominated directors

#3

Target mid-range budgets rather than blockbuster budgets

Executive Recommendations



The Investor

#1

Commercial over acclaim

#2

Manage production costs carefully

#3

Diversify the portfolio

THANK YOU

